

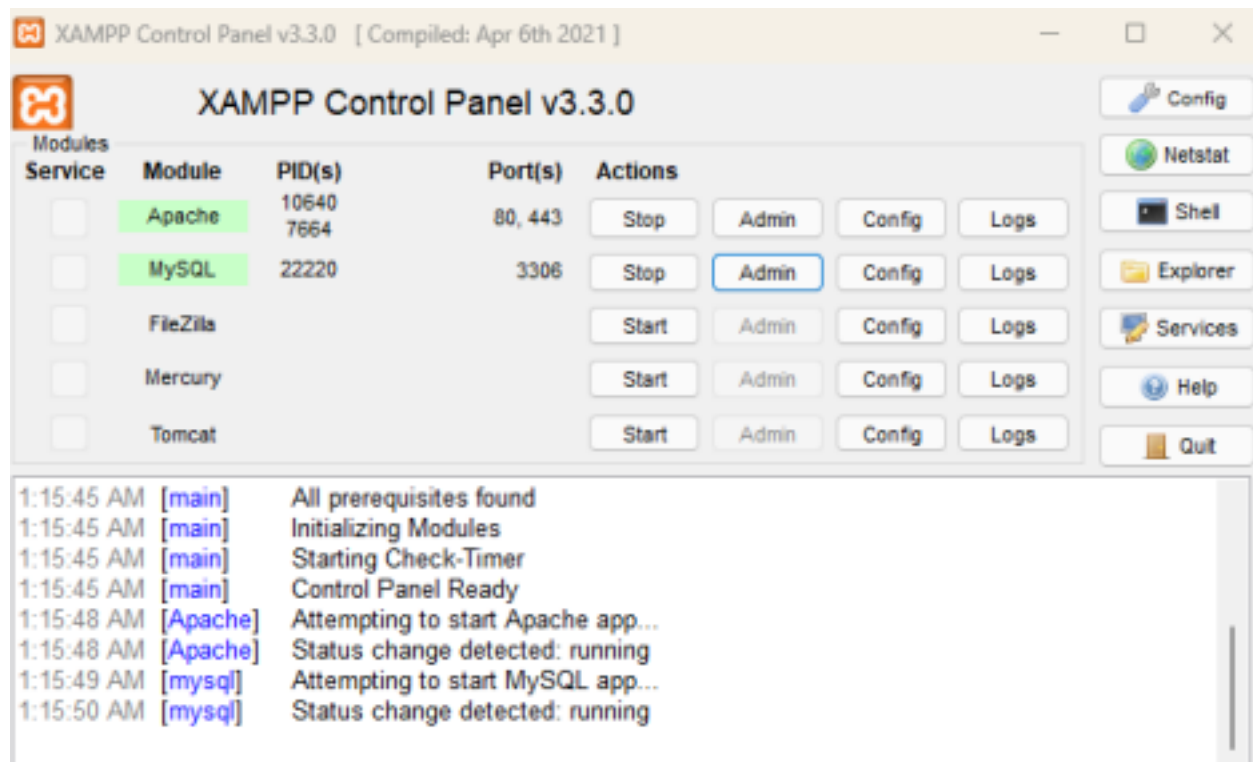
C1A: Coding & MySQL Integration

Objective: Implement CODING scripts to interact with a MySQL database, reinforcing database connectivity, CRUD operations, and data retrieval.

Instructions:

Steps:

- a) Install Xampp
- b) Start Apache and MySQL

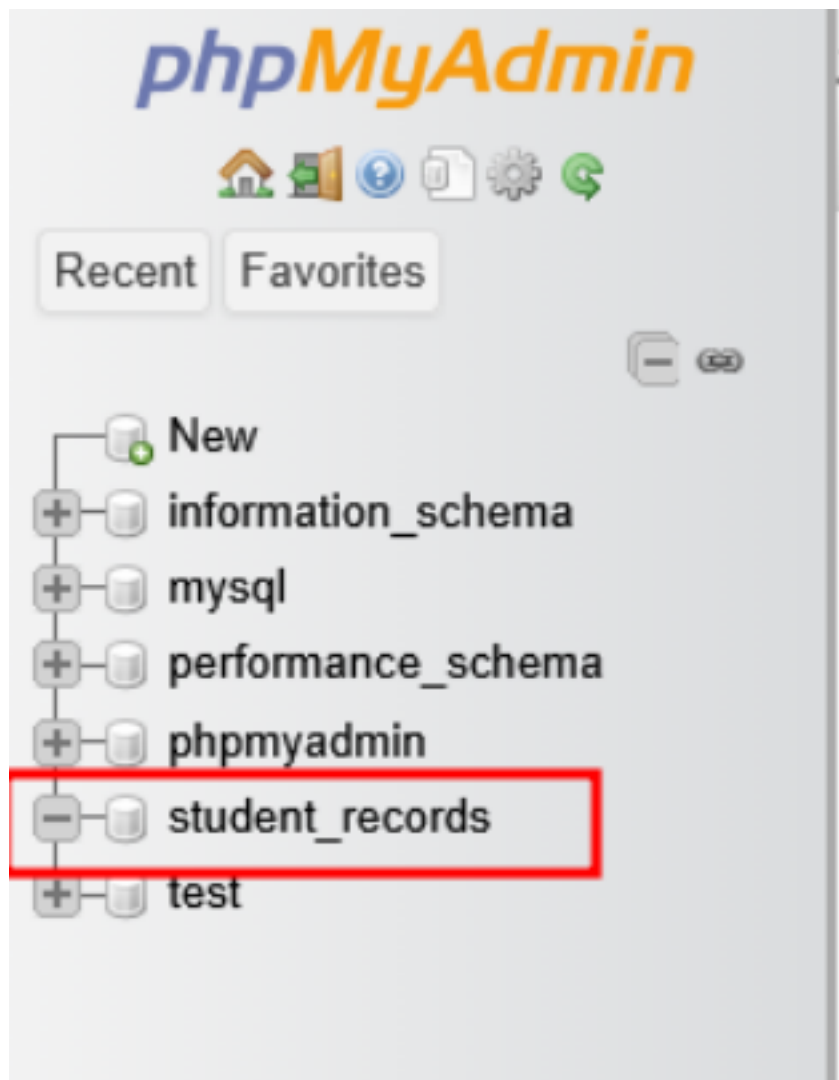


- c) Open <http://localhost/phpmyadmin> on browser, where to interact with databases.

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1. Database Creation:

Using MySQL, create a database named **student_records**.



2. Table Creation:

Within the student_records database, create a table called students with the following columns:

- id (INT, Primary Key, Auto Increment)
- name (VARCHAR(100))

- age (INT)
- email (VARCHAR(100))

Table name: Add

Structure

Name	Type	Length/Values	Default	Collation	Attributes	Null Index	Comments	Virtuality	More columns
id	INT		None			☒	PRIMARY		
name	VARCHAR	100	None			☐			
age	INT		None			☐			
email	VARCHAR	100	None			☐			

Table comment:

Collation:

Storage Engine:

PARTITION definition

Partition by:

Partitions:

Server: 127.0.0.1 > Database: student_records > Table: students

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	☒ id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	age	int(11)			No	None			Change Drop More
☐ 4	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

3. Data Insertion:

Insert at least three records into the students table.

```
INSERT INTO `students` (`name`, `age`, `email`)
VALUES
('Benson', '19', 'makaubenson@mail.com'),
('Shanice', '21', 'shanice.smith@example.com'),
('Emilio', '21', 'emilio.smith@example.com'),
('Emilio', '21', 'emilio.smith@example.com'),
```

```
('Elvis', '20', 'elvis.jones@example.com');
```

The top screenshot shows the phpMyAdmin interface with the SQL tab selected. The query being executed is:

```

1 INSERT INTO `students` (`name`, `age`, `email`)
2 VALUES
3 ('Benson', '19', 'makaubenson@mail.com'),
4 ('Shanice', '21', 'shanice.smith@example.com'),
5 ('Emilio', '21', 'emilio.smith@example.com'),
6 ('Emilio', '21', 'emilio.smith@example.com'),
7 ('Elvis', '20', 'elvis.jones@example.com');

```

The bottom screenshot shows the result of the query, displaying the five records in a table format. The table has columns: id, name, age, email. The records are:

id	name	age	email
1	Benson	19	makaubenson@mail.com
2	Shanice	21	shanice.smith@example.com
3	Emilio	21	emilio.smith@example.com
4	Emilio	21	emilio.smith@example.com
5	Elvis	20	elvis.jones@example.com

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4. Data Retrieval:

Write a CODING script to connect to the MySQL database and retrieve all records from the students table. Display the records in an HTML table.

a) PHP Script

Student Records

#	Name	Age	Email
1	Benson	19	makaubenson@mail.com
2	Shanice	21	shanice.smith@example.com
3	Emilio	21	emilio.smith@example.com
4	Emilio	21	emilio.smith@example.com
5	Elvis	20	elvis.jones@example.com

5. Data Update:

Implement a CODING script that updates a student's email based on their id.

Update Student Email

Update Email by Student ID

Student ID

New Email Address

Update Email

Home

Current Student Records

ID	Name	Age	Email
1	Benson	19	makaubenson@gmail.com
2	Shanice	21	shanice.smith@example.com
3	Emilio	21	emilio.smith@example.com
4	Emilio	21	emilio.smith@example.com
5	Elvis	20	elvis.test@example.com

Update Student Email

Update Email by Student ID

Student ID

1

New Email Address

makaubenson@africahackon.com

Update Email Home

Current Student Records

ID	Name	Age	Email
1	Benson	19	makaubenson@gmail.com
2	Shanice	21	shanice.smith@example.com
3	Emilio	21	emilio.smith@example.com
4	Emilio	21	emilio.smith@example.com
5	Elvis	20	elvis.test@example.com

Update Student Email

Email successfully updated for student: Benson

Update Email by Student ID

Student ID

New Email Address

Update Email Home

Current Student Records

ID	Name	Age	Email
1	Benson	19	makaubenson@africahackon.com
2	Shanice	21	shanice.smith@example.com
3	Emilio	21	emilio.smith@example.com
4	Emilio	21	emilio.smith@example.com
5	Elvis	20	elvis.test@example.com


```

update_email.php
1 <body>
2 <div class="container mt-5">
3
4
5
6
7
8
9
10
11 $message = '';
12 $messageType = '';
13
14 // Handle form submission
15 if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['update_email'])) {
16     $studentId = $_POST['student_id'];
17     $newEmail = $_POST['new_email'];
18
19     try {
20         $pdo = new PDO($data_source, $user, $pass, $options);
21
22         // First, check if the student exists
23         $checkSql = "SELECT 'name' FROM 'students' WHERE 'id' = :id";
24         $checkStat = $pdo->prepare($checkSql);
25         $checkStat->execute(['id' => $studentId]);
26         $student = $checkStat->fetch();
27
28         if ($student) {
29             // Update the email
30             $updateSql = "UPDATE 'students' SET 'email' = :email WHERE 'id' = :id";
31             $updateStat = $pdo->prepare($updateSql);
32             $updateStat->execute([
33                 'email' => $newEmail,
34                 'id' => $studentId
35             ]);
36
37             $message = "Email successfully updated for student: " . htmlspecialchars($student['name']);
38             $messageType = 'success';
39         } else {
40             $message = "Student with ID $studentId not found.";
41             $messageType = 'warning';
42         }
43     } catch (PDOException $e) {
44         $message = "Database Error: Could not update email.";
45         $messageType = 'danger';
46     }
47 }
48
49 // Display message if exists
50 if ($message) {
51     echo "<div class='alert alert-$messageType' role='alert'>$message</div>";
52 }
53 }>
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70

```

8

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6. Data Deletion:

Create a CODING script to delete a student record based on their id.

https://localhost/coding-ah/delete_student.php

Delete Student Record

Current Student Records

ID	Name	Age	Email	Action
1	Benson	19	makaubenson@afrihackon.com	Delete
2	Shanice	21	shanice.smith@example.com	Delete
3	Emilio	21	emilio.smith@example.com	Delete
4	Emilio	21	emilio.smith@example.com	Delete
5	Evis	20	evis.test@example.com	Delete



